

Figure S1. Distribution of ZIP code-level study population size 2011-2016

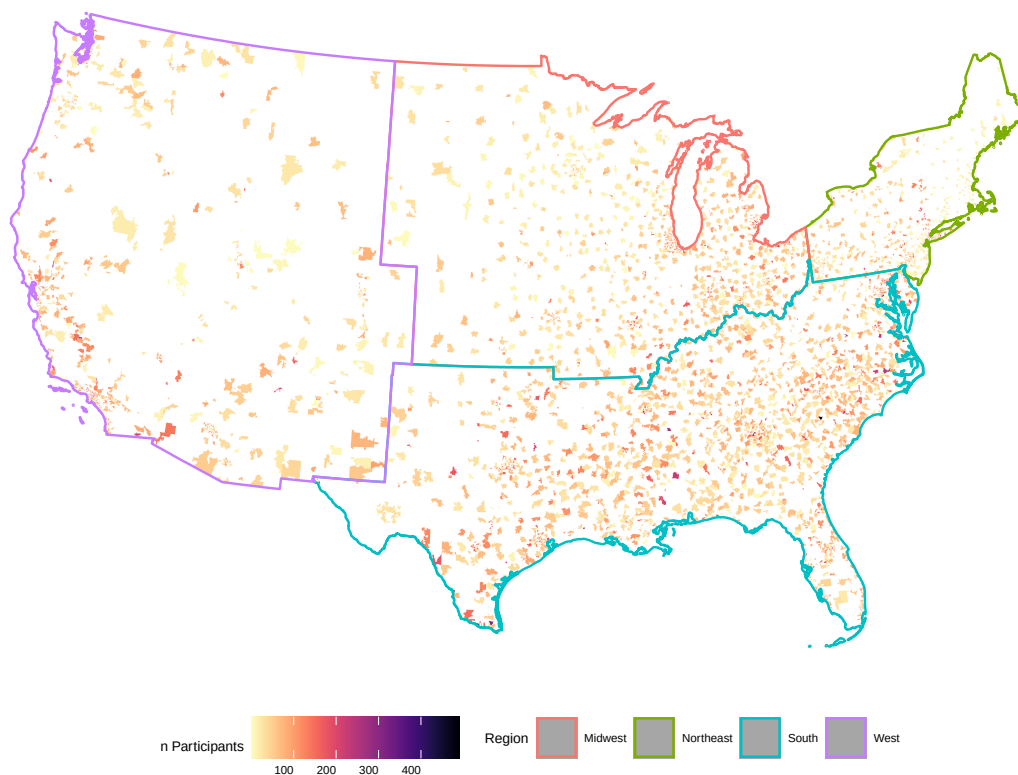
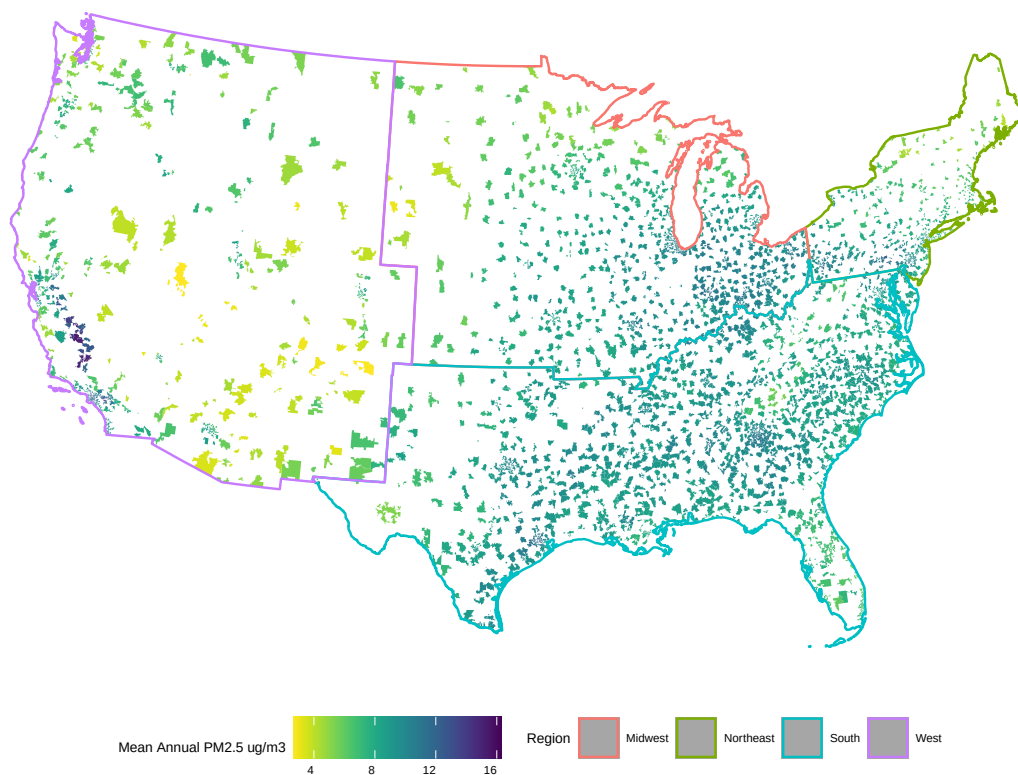


Figure S2. Distribution of ZIP code-level average annual- PM_{2.5} 2011-2016.



Item S1. Model specification

We incorporated time since the initiation of dialysis as the analysis time scale to control for the potential temporal effect variation by this factor, which has been suggested to be a predictor of mortality and comorbidities among dialysis patients.^{1,2} Daily temperature (linear), daily relative humidity (linear), DOW, season (spring, summer, fall, and winter) were included to control for potential time varying confounding. Specifically, DOW and season were incorporated to account for short- and long-term temporal variability of the outcomes.^{3,4} Time-invariant confounders included age at dialysis initiation, race (American Indian/Native America, Asian, Black, Hispanic, Pacific Islander, White, Other), baseline employment status, and geographic region (Northeast, Midwest, South, and West; see below).

Geographic Region Definition.

Region	States
<i>Northeast</i>	Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont, New Jersey, New York, Pennsylvania
<i>Midwest</i>	Illinois, Indiana, Michigan, Ohio, Wisconsin, Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota
<i>South</i>	Delaware, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, Washington, D.C., West Virginia, Alabama, Kentucky, Mississippi, Tennessee, Arkansas, Louisiana, Oklahoma, Texas
<i>West</i>	Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, Wyoming, California, Oregon, Washington

Baseline employment status was incorporated as a surrogate for socio-economic status (SES) as classical SES variables like income or education level information were not available.

Potential effect measure modifications by baseline comorbidity statuses, race, sex, and dialysis initiating age were assessed. The comorbidities assessed were diabetes, CVD, and chronic obstructive pulmonary disease (COPD) reported at the start of maintenance hemodialysis on the CMS 2728 form (Table S2). Likelihood ratio tests for homogeneity were performed by comparing models with and without an interaction term between the potential modifier and PM

term. A small p-value (less than 0.05) from Chi-square test of likelihood ratios test was used to reject the null hypothesis of homogeneity, indicating PM effects are not equivalent across different levels of potential modifiers.

Data management and statistical analysis were performed using SAS software, Version 9.4 (SAS Institute Inc., Cary, NC, USA). Plots were generated with R Project for Statistical Computing Version 3.5.0 (R Foundation for Statistical Computing, Vienna, Austria).

References

1. Chertow GM, Johansen KL, Lew N, Lazarus JM, Lowrie EG. Vintage, nutritional status, and survival in hemodialysis patients. *Kidney international*. 2000;57(3):1176-1181.
2. Sumida K, Yamagata K, Iseki K, Tsubakihara Y. Different impact of hemodialysis vintage on cause-specific mortality in long-term hemodialysis patients. *Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association*. 2016;31(2):298-305.
3. Peng RD, Dominici F, Pastor-Barriuso R, Zeger SL, Samet JM. Seasonal analyses of air pollution and mortality in 100 US cities. *American journal of epidemiology*. 2005;161(6):585-594.
4. Zhang H, Schaubel DE, Kalbfleisch JD, et al. Dialysis outcomes and analysis of practice patterns suggests the dialysis schedule affects day-of-week mortality. *Kidney international*. 2012;81(11):1108-1115.

Table S1. Cause of death and ICD codes used to identify CV events.

Event	Codes		Count (%) in First CV Event Analysis
CVD related mortality	Primary Cause of Death*		4,143 (0.67)
	23, 25, 26, 27, 28, 29, 30, 31, 32, 35, 36		
CVD related ED visits and hospital admission	ICD-9	ICD-10	
<i>Hypertensive disease</i>	402, 404	I11, I13	18,712 (3.01)
<i>Ischemic heart disease</i>	410, 411, 412, 413, 414	I20, I21, I22, I23, I24, I25.	170,610 (27.48)
<i>Heart failure, cardiac arrest, and related</i>	398, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429	I09, I30, I31, I32, I33, I34, I35, I36, I37, I38, I39, I40, I41, I42, I43, I44, I45, I46, I47, I48, I49, I50, I51.	370,925 (59.75)
<i>Cerebrovascular</i>	430, 431, 432, 433, 434, 435, 436, 437, 438	I60, I61, I62, I63, I64, I65, I66, I67, I68, I69, G45, G46.	56.357 (9.08)

* The primary cause of death codes are based on the CMS 2746 form list of causes.

Table S2. Baseline comorbidity status classification.

Baseline Comorbidity	If “yes” to any of the following CMS 2728 questions
CVD	atherosclerotic heart disease, cardiac arrest, congestive heart failure, cerebrovascular disease, cardiac dysrhythmia, ischemic heart disease, myocardial infarction, pericarditis, peripheral vascular disease, and other cardiac disease
DM	diabetes (primary or contributing), diabetic and currently on insulin, on oral medications, or without medications, or with diabetic retinopathy
COPD	chronic obstructive pulmonary disease

Table S3. United States, 2011-2016 ZIP code-level daily meteorological variables distribution.

A. Mortality analysis (n = 193,121,928 person-days)

	Mean (STD)	Range	Median	Q1	Q3
Temperature (°C)	15.6 (9.7)	-34.7, 41.5	17.1	9.0	23.3
Relative Humidity (%)	65.7 (15.7)	0, 121.0	67.7	56.7	76.8

B. CV event analysis (n = 94,385,773 person-days)

	Mean (STD)	Range	Median	Q1	Q3
Temperature (°C)	15.6 (9.7)	-31.5, 41.5	17.1	9.1	23.3
Relative Humidity (%)	65.4 (15.9)	0, 121.0	67.4	56.3	76.6

Table S4. CV event, CVD-specific mortality, and all-cause mortality overall and stratification analysis by age at dialysis initiation, sex, race ethnicity, vascular access type, and comorbidity status at dialysis initiation. Estimates (HRs per 1 $\mu\text{g}/\text{m}^3$ increase) of association between annual average PM_{2.5} exposure and all-cause mortality among HD patients, 2011-2016

Cohort	CVD Event	CVD-Specific Mortality	All-Cause Mortality
Age at Dialysis Initiation			
18-44	1.011 (1.002, 1.020)	0.995 (0.968, 1.022)	1.001 (0.983, 1.020)
45-64	1.015 (1.010, 1.019)	1.006 (0.995, 1.016)	1.016 (1.008, 1.023)
65-74	1.016 (1.011, 1.021)	1.019 (1.008, 1.031)	1.020 (1.012, 1.027)
75-over	1.021 (1.016, 1.025)	1.044 (1.034, 1.054)	1.037 (1.031, 1.044)
Sex			
Female	1.017 (1.013, 1.020)	1.019 (1.010, 1.029)	1.022 (1.016, 1.028)
Male	1.016 (1.013, 1.019)	1.025 (1.017, 1.033)	1.026 (1.020, 1.031)
Race Ethnicity			
American Indian/Native America	1.010 (0.990, 1.030)	0.998 (0.947, 1.053)	0.992 (0.958, 1.027)
Asian	1.014 (1.001, 1.027)	1.063 (1.029, 1.098)	1.052 (1.029, 1.076)
Black	1.008 (1.002, 1.013)	0.982 (0.968, 0.997)	1.016 (1.006, 1.026)
Hispanic	1.012 (1.006, 1.017)	1.022 (1.007, 1.037)	1.023 (1.013, 1.033)
Pacific Islander	1.014 (0.987, 1.042)	0.977 (0.913, 1.045)	0.976 (0.929, 1.024)
White	1.020 (1.017, 1.024)	1.032 (1.024, 1.040)	1.026 (1.021, 1.032)
Other	0.965 (0.887, 1.051)	0.941 (0.775, 1.143)	0.954 (0.848, 1.073)
Baseline DM			
yes	1.016 (1.013, 1.019)	1.017 (1.010, 1.025)	1.020 (1.015, 1.025)
no	1.017 (1.013, 1.021)	1.030 (1.020, 1.040)	1.028 (1.022, 1.035)
Baseline CVD			
yes	1.016 (1.013, 1.019)	1.023 (1.016, 1.031)	1.023 (1.022, 1.035)
no	1.019 (1.016, 1.023)	1.026 (1.015, 1.036)	1.028 (1.022, 1.035)
Baseline COPD			
yes	1.024 (1.017, 1.032)	1.026 (1.009, 1.043)	1.024 (1.013, 1.035)
no	1.016 (1.014, 1.019)	1.023 (1.016, 1.030)	1.025 (1.021, 1.029)
Overall	1.016 (1.014, 1.019)	1.022 (1.016, 1.028)	1.024 (1.020, 1.028)

Table S5. CV event specification sensitivity analysis.

Outcome Definition	Beta	SE	HR (95%CI) per 8.7 µg/m³
All CV (reported as main results)	0.0162	0.00123	1.15 (1.13, 1.18)
All CV excluding event with only hypertensive codes	0.0159	0.00124	1.15 (1.12, 1.17)

Table S6. Study cohort age at dialysis initiation distribution (%) by potential modifiers (baseline comorbidities, race ethnicity, and gender).

	<i>Age Categories</i>			
	<i>18-44</i>	<i>45-64</i>	<i>65-74</i>	<i>75-older</i>
Overall Study Population	10.9	37.9	26.0	25.2
Baseline CVD				
Yes	5.2	34.0	29.3	31.4
No	17.1	42.3	22.4	18.2
Baseline DM				
Yes	8.6	41.8	28.3	21.4
No	14.1	32.4	22.8	30.6
Baseline COPD				
Yes	2.2	31.0	34.3	32.6
No	11.8	38.7	25.1	24.3
Race Ethnicity				
American Indian/ Alaskan Native	15.2	49.9	23.1	11.8
Asian	9.9	32.3	26.0	31.8
Black	17.1	45.1	22.1	15.8
Hispanic	13.9	49.5	21.6	15.0
Pacific Islander	12.9	46.6	23.4	17.0
White	6.6	30.9	29.5	33.1
Other	88.3	4.0	4.2	3.4
Gender				
Male	11.5	39.6	24.8	24.1
Female	10.1	35.6	27.7	26.7